SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE – AI	**Date:**	19-08-2026

Code of Conduct

I N. Tharun Sai *(192472382) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I Understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: N. Tharun Sai

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether **"to the car"** is associated with *reported* or *damage*.

Tasks:

a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.

b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.

c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.

d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

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2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates **"with AC sleeper"** with the passengers instead of the ticket/class information.

Tasks:

a) Analyze the possible syntactic interpretations of the given passenger request using CFG.

b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.

c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.

d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

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3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved

- Obligations
- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

a) Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.

b) Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.

c) Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.

d) Develop a step-by-step processing workflow showing how the system can extract:

- Supplier
- Action
- Equipment
- Company
- Obligation
- Deadline

and justify how the proposed approach improves automated contract analysis.

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4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate **"for students"** with the touchscreen rather than the laptop.

Tasks:

a) Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.

b) Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.

c) Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.

d) Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

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5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, **"with two passengers"** may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

a) Analyze the possible syntactic structures and ambiguities in the given voice command.

b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.

c) Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.

d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

1. AI-Powered Insurance Claim Processing System

a) Structural ambiguity using CFG

Sentence:

"I reported the damage to the car after the accident."

The phrase "to the car" creates PP attachment ambiguity.

Interpretation 1: "to the car" modifies *damage*.

I reported the damage to the car after the accident.

Interpretation 2: "to the car" is attached to the verb phrase *reported*.

A simple CFG can be:

S → NP VP
VP → V NP PP
NP → Det N
NP → NP PP
PP → P NP

Thus, CFG can generate more than one valid parse tree for the same sentence.

b) Limitations of basic CFG

- CFG cannot easily resolve structural ambiguity.
- It does not naturally handle subject–verb agreement.
- Long sentences can produce many parse trees.
- It performs poorly with incomplete conversational input.
- It does not assign probabilities to different interpretations.

c) Improved parsing approach

The system can combine:

- CFG – provides basic grammatical rules.
- Earley Parsing – handles ambiguity, incomplete input, and long sentences.
- PCFG – assigns probabilities and selects the most likely parse.
- Feature Structures – handle features such as number, person, tense, and semantic roles.

Flow:

Customer Input
↓
Preprocessing
↓
Earley Parser
↓
PCFG Ranking
↓
Feature Checking
↓
Best Parse
↓
Claim Information

d) Benefits

The proposed system provides:

- Better ambiguity resolution
- Improved grammatical accuracy
- Faster processing of complex sentences
- Better handling of conversational input
- More reliable extraction of claimant, object, cause, and amount

2. Intelligent Railway Ticket Booking Assistant
a) CFG interpretation
Sentence:
"Book two tickets from Chennai to Delhi for my parents with AC sleeper."
The intended interpretation is:
Book two tickets from Chennai to Delhi, for my parents, with AC sleeper class.
The phrase "with AC sleeper" can also be incorrectly attached to "my parents", creating ambiguity.
A simple CFG can be:
S → VP
VP → V NP
NP → NP PP
PP → P NP
b) Problems with Top-Down Parsing
Top-down parsing starts from the start symbol and predicts the structure.
Problems include:
- Excessive backtracking
- Repeated parsing of the same input
- Difficulty with ambiguous phrases
- Poor handling of incomplete requests
- Slow processing for long queries

c) Earley Parsing
Earley Parsing maintains different possible parse states while processing the sentence.
It can:
- Handle ambiguous structures
- Process incomplete input
- Avoid unnecessary repeated parsing
- Handle long and complex queries

Therefore, it is suitable for conversational ticket booking.
d) Comparison

Feature	Top-Down	CFG	Earley
Ambiguity	Weak	Generates alternatives	Handles alternatives
Backtracking	High	Depends on parser	Reduced
Incomplete input	Poor	Limited	Good
Complex queries	Less efficient	Moderate	Better
Real-time use	Limited	Moderate	Suitable

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3. AI-Based Legal Contract Analysis System

a) CFG structure

Sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The main structure is:

S → NP VP

NP → NP RelativeClause

VP → V NP PP

The relative clause "who delivered the equipment to the company" describes the supplier.

The main obligation is:

Supplier → replace defective components → within 30 days.

b) Limitations of basic CFG

Basic CFG has difficulty with:

- Relative clauses
- Nested structures
- Passive constructions
- Multiple prepositional phrases
- Long legal sentences
- Identifying semantic relationships

It can identify grammar but cannot reliably determine who has an obligation.

c) Improved architecture

Legal Sentence

↓

CFG

↓

Earley Parsing

↓

PCFG Ranking

↓

Feature Structures

↓

Semantic Analysis

↓

Contract Information

- CFG: basic grammar
- Earley: handles complex and ambiguous structures
- PCFG: selects the most probable interpretation
- Feature Structures: maintain grammatical and semantic information

d) Information extraction

Information	Value
Supplier	The supplier
Action	Replace
Equipment	The equipment
Company	The company
Obligation	Replace defective components
Deadline	Within 30 days

This approach improves the accuracy of automated contract analysis.

4. AI-Based E-Commerce Product Search System

a) CFG ambiguity

Query:
"Find laptops with a touchscreen for students under ₹60,000."
Intended meaning:
Find laptops having a touchscreen, suitable for students, costing below ₹60,000.
The phrase "for students" can incorrectly be attached to "touchscreen" instead of "laptops."
Simple CFG:
S → VP
VP → V NP
NP → N PP
NP → NP PP
PP → P NP

b) Limitations of CFG

CFG has difficulty handling:

- Ambiguous phrase attachment
- Informal queries
- Missing words
- Coordination
- Long descriptions
- Real-time search requirements

Example:
"Laptop touchscreen under 60k"
is incomplete but should still be understood.

c) Improved architecture

User Query
↓
Earley Parsing
↓
PCFG Ranking
↓
Feature Structures
↓
Search Constraints
↓
Product Results
Feature structure:
Product = Laptop
Feature = Touchscreen
User = Students
Price ≤ ₹60,000

d) Benefits

The proposed approach provides:

- Better search precision
- Faster interpretation
- Correct identification of product attributes
- Support for natural-language queries
- Better customer experience

5. Intelligent Airline Voice Assistant

a) Syntactic ambiguity

Command:

"Change my flight to Mumbai tomorrow with two passengers."

Intended interpretation:

Change my flight to Mumbai tomorrow, for two passengers.

The phrase "with two passengers" may be incorrectly attached to "Mumbai."

The correction:

"Mumbai tomorrow... actually, make that Bangalore."

means the destination should be changed from Mumbai to Bangalore.

b) Limitations of Top-Down Parsing

Top-down parsing may suffer from:

- Excessive backtracking
- Slow response
- Difficulty handling incomplete speech
- Problems with speech recognition errors
- Difficulty handling corrections
- Ambiguous phrase attachment

c) Improved architecture

Voice Input

↓

Speech Recognition

↓

Earley Parsing

↓

PCFG Ranking

↓

Feature Structures

↓

Semantic Information

↓

Booking Request

Earley Parsing handles partial and ambiguous input.

PCFG selects the most probable interpretation.

Feature Structures maintain information such as destination, date, and number of passengers.

d) Processing workflow

Input:

"Change my flight to Mumbai tomorrow with two passengers."

Extracted information:

Feature	Value
Action	Change Flight
Destination	Mumbai
Date	Tomorrow
Passengers	2

After correction:

"Actually, make that Bangalore."

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1	3	3	4	4	4		18	88
2	3	3	4	4	4		18	
3	4	4	3	3	4		18	
4	4	4	3	3	4		18	
5	4	4	4	3	3		18	

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